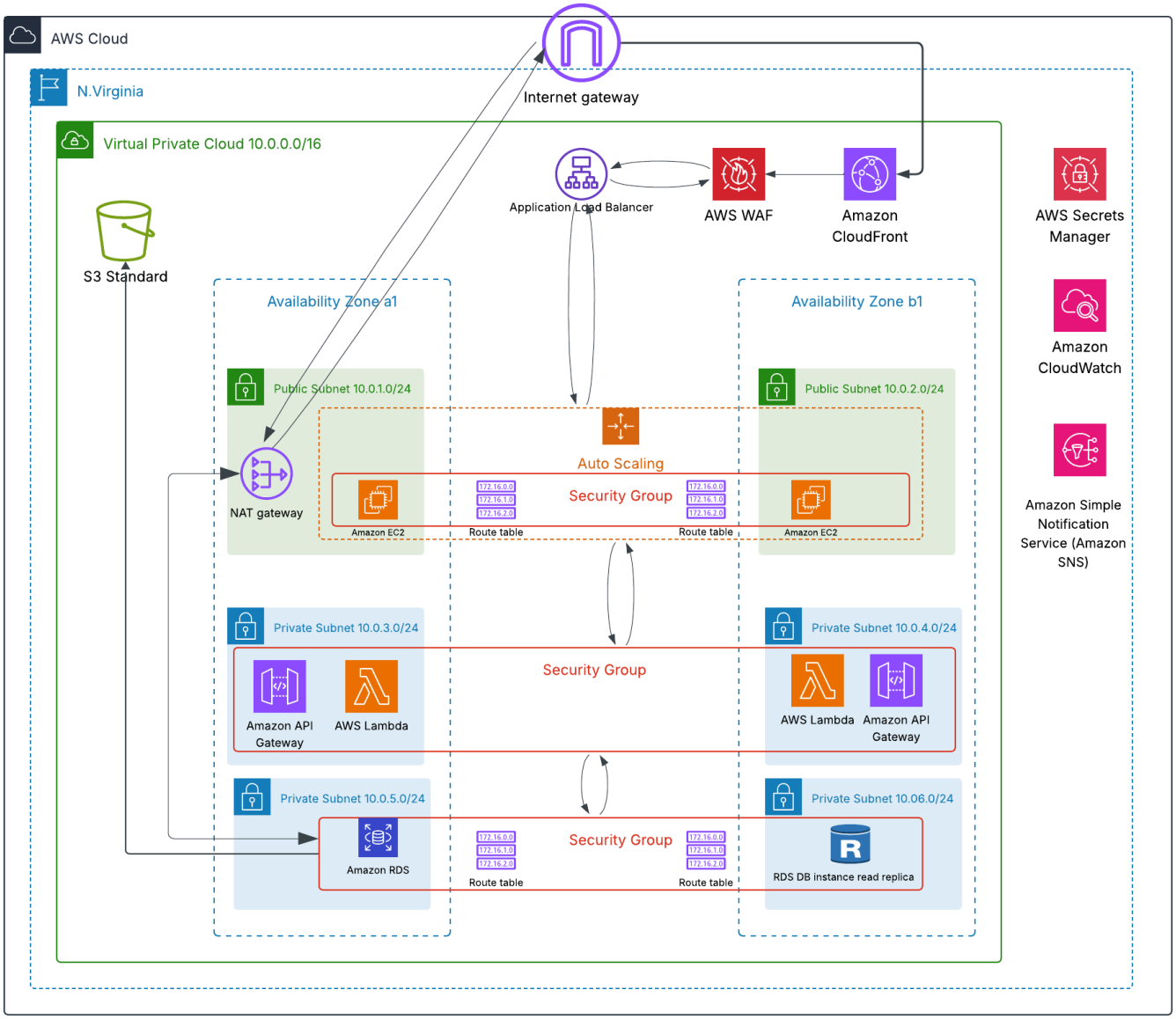
Cloud-Based Healthcare Management System

Introduction:-

I designed a cloud-based healthcare management system on AWS in this project. This system enables patients to schedule appointments, access medical records, and securely communicate with healthcare providers. Additionally, it includes features for doctors to manage appointments, prescriptions, and patient medical histories. The architecture is highly scalable, secure, and optimized for performance.

Architecture Diagram:



AWS Services Used

1. **Amazon CloudFront**

    - Distributes content globally with low latency and improves static and dynamic content delivery performance.

2. **AWS WAF (Web Application Firewall)**

    - Protects the application from malicious attacks like SQL injection and cross-site scripting.

3. **Application Load Balancer (ALB)**

    - Distributes incoming traffic across multiple EC2 instances in different Availability Zones for high availability.

4. **Auto Scaling**

    - Automatically adjusts the number of EC2 instances to handle changes in traffic demand, ensuring cost-efficiency and reliability.

5. **Amazon EC2**

    - Hosts the application frontend and backend services in a scalable and reliable environment.

6. **<u>Amazon RDS (Relational Database Service)</u>**
   - Provides a secure and highly available database for storing patient records and medical data.

7. **<u>AWS Lambda</u>**

   - Handles serverless backend logic, triggered by API Gateway, to process patient and doctor requests efficiently.

8. **<u>Amazon S3</u>**

   - Stores static assets such as images, CSS, and JavaScript files for the web application.

9. **<u>Amazon SNS (Simple Notification Service)</u>**

   - Sends notifications to patients and doctors for appointment updates, reminders, or other important alerts.

10. **<u>Amazon CloudWatch</u>**

   - Monitors application performance, providing insights into EC2, RDS, and Lambda metrics, and triggers alarms when necessary.

How the Architecture Works

1. **Frontend Layer:**

   - Users (patients and doctors) access the application through a browser or mobile device.
   - Static assets are delivered via Amazon CloudFront, improving performance for global users.
   - Dynamic requests are routed through AWS WAF for security and forwarded to the Application Load Balancer (ALB).

2. **Backend Layer:**

   - The ALB routes incoming requests to an Auto Scaling group of EC2 instances hosting the application.
   - For serverless requests, API Gateway triggers AWS Lambda, which handles backend logic.
   - Both EC2 and Lambda access the database stored in Amazon RDS.

3. **Database Layer:**

   - Amazon RDS is hosted in private subnets across multiple Availability Zones, ensuring high availability and data security.

4. **Storage and Notifications:**

- Static files are stored in Amazon S3, and Amazon SNS sends appointment notifications and alerts to users.

5. **Monitoring and Security:**

- AWS CloudWatch tracks application performance and sends alarms for unusual activity.
- AWS Cognito ensures secure user authentication, and WAF protects against malicious traffic.

Conclusion

This architecture ensures a highly available, scalable, and secure solution for managing healthcare needs. By leveraging AWS services, the system is designed to handle varying traffic loads, protect sensitive medical data, and provide seamless communication between patients and doctors. It lays a solid foundation for future scalability and additional features.